

Answer of 01

Given expressions,

$$(A'B+B')BC'C + CB'C + D$$

$$= (A'B+B')B \cdot CC' + C \cdot C \cdot B' + D$$

$$= (A'B+B')B \cdot 0 + CB' + D \quad \left[\text{Here, } \begin{matrix} CC' = 0 \\ C \cdot C = C \end{matrix} \right]$$

$$= 0 + CB' + D$$

$$= B'C + D$$

$\therefore$ Simplified expression of the given expression is $B'C + D$

Ans

Answer of 02

Given expression,

$$F = (x' + y + z')(x' + y')(x + z')$$

Complement of the expression,

$$F' = [(x' + y + z')(x' + y')(x + z')]'$$

$$= (x' + y + z')' + (x' + y')' + (x + z')'$$

$$= [x' \cdot y' \cdot (z')'] + [x' \cdot (y)'] + [x \cdot (z)']$$

$$= x'yz + xy + xz$$

Ans

Answer of 03

SOP

$$F(A, B, C, D) = A + B'CD'$$

$$= A(B+B)(C+C)(D+D) + B'CD'(A+A)$$

$$= (AB+AB')(C+C)(D+D) + AB'CD' + A'B'CD'$$

$$= (ABC+ABC'+ABC'+ABC'')(D+D) + AB'CD' + A'B'CD'$$

$$= ABCD + ABCD' + ABC'D + ABC'D' + A'B'CD + A'B'CD' + A'B'CD + A'B'CD'$$

$$1111 + 1110 + 1101 + 1100 + 1011 + 1010 + 1001 +$$

$$1000 + 0010$$

$$= \sum (15, 14, 13, 12, 11, 10, 9, 8, 2)$$

$$= \sum (2, 8, 9, 10, 11, 12, 13, 14, 15)$$

POS

$$F(A, B, C, D) = A + B'CD'$$

$$= (A+B')(A+CD')$$

$$= (A+B')(A+C)(A+D')$$

$$= (A+B'+CC')(A+C+BB')(A+D'+BB')$$

$$= (A+B'+C)(A+B'+C')(A+C+B)(A+C+B')(A+D'+B)(A+D'+B')$$

$$= (A+B'+C+DD')(A+B'+C'+DD')(A+C+B+DD')(A+C+B'+DD')(A+D'+B+CC')(A+D'+B'+CC')$$

$$= (A+B'+C+D)(A+B'+C'+D)(A+B'+C+D')(A+B'+C'+D')(A+B'+C+D)(A+B'+C'+D)(A+B'+C+D)(A+B'+C'+D)$$

$$= (0+1+0+0)(0+1+0+1)(0+1+1+0)(0+1+1+1)(0+0+0+0)(0+0+0+1)(0+1+0+0)(0+1+0+1)(0+0+0+1)(0+0+0+1)(0+1+0+0)(0+1+1+0)$$

$$= \prod (0, 1, 3, 4, 5, 6, 7) \text{ Ans}$$

SOP

$$F(A, B, C, D, E) = AB + CDE$$

$$AB(C+C')(D+D')(E+E') + CDE(A+A')(B+B')$$

$$= (ABC + ABC')(D+D')(E+E') + (ACDE + AC'DE)(B+B')$$

$$= (ABCD + ABC'D + ABCD' + ABC'D')(E+E') + ABCDE + ABC'DE + ABCDE' + ABC'DE'$$

$$= ABCDE + ABC'DE + ABCDE' + ABC'DE' + ABCDE' + ABC'DE' +$$

$$+ ABC'DE' + ABC'DE' + ABCDE + ABC'DE + ABCDE + ABC'DE$$

$$= \underline{11111} + 11011 + 11101 + 11001 + 11110 + 11010 +$$

$$11100 + 11000 + \underline{11111} + 01111 + 10111 + 00111$$

$$= \Sigma(31, 27, 29, 25, 30, 26, 28, 24, 15, 23, 7)$$

$$= \Sigma(7, 15, 23, 24, 25, 26, 27, 28, 29, 30, 31)$$

POS

$$F(A, B, C, D, E) = AB + CDE$$

$$= (AB+C)(AB+DE)$$

$$= (A+C)(B+C)(A+D)(A+E)$$

$$= (A+C)(B+C)(A+D)(B+D)(A+E)(B+E)$$

$$= (A+C+BB')(B+C+AA')(A+D+BB')(B+D+AA')(A+E+BB')(B+E+AA')$$

$$= (A+B+C)(A+B'+C)(A+B+C)(A'+B+C)(A+B+D)(A+B'+D)(A+B+D)(A'+B+D)(A+B+E)$$

$$(A+B'+E)(A+B'+E)(A'+B+E)$$

$$= (A+B+C+DD')(A+B'+C+DD')(A+B+C+DD')(A'+B+C+DD')(A+B+D+CC')(A+B'+D+CC')$$

$$(A+B+D+CC')(A'+B+D+CC')(A+B+E+CC')(A+B'+E+CC')(A+B+E+CC')(A'+B+E+CC')$$

$$= (A+B+C+D)(A+B+C+D')(A+B'+C+D)(A+B'+C+D')(A+B+C+D)(A+B'+C+D)(A'+B+C+D)(A'+B+C+D')$$

$$(A+B+C+D)(A+B'+C+D)(A+B'+C+D)(A+B'+C+D)(A+B+C+D)(A+B'+C+D)(A'+B+C+D)(A'+B+C+D')$$

$$(A'+B+C+D)(A+B+C+E)(A+B'+C+E)(A+B'+C+E)(A+B'+C+E)(A+B'+C+E)(A+B'+C+E)(A+B'+C+E)$$

$$(A'+B+C+E)(A'+B+C+E)$$

$$\begin{aligned}
 &= (A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE') \\
 &\quad (A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE') \\
 &\quad (A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE') \\
 &\quad (A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE') \\
 &\quad (A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE') \\
 &\quad (A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE') \\
 &\quad (A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')(A+B+C+D+EE')
 \end{aligned}$$

$$\begin{aligned}
 &= (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &\quad (A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E)(A+B+C+D+E) \\
 &= \prod_{i=0,1,2,3,4,5,6,8,9,10,11,12,13,14,16,17,18,19,20,21,22} (A+B+C+D+E)
 \end{aligned}$$

Answer of Q4

Given equation,

$$F(a,b,c,d) = \sum(8,9,10,11,13,15)$$

Binary

$$8 \rightarrow 1000 \rightarrow ab'c'd'$$

$$9 \rightarrow 1001 \rightarrow ab'c'd$$

$$10 \rightarrow 1010 \rightarrow ab'cd'$$

$$11 \rightarrow 1011 \rightarrow ab'cd$$

$$13 \rightarrow 1101 \rightarrow abc'd$$

$$15 \rightarrow 1111 \rightarrow abcd$$

SOP Form

$$F = ab'c'd' + ab'c'd + ab'cd' + ab'cd + abc'd + abcd$$

$$= ab'(c'd' + c'd + cd' + cd) + ab(c'd + cd)$$

$$= ab'(c'(d'+d) + c(d'+d)) + ab(d(c'+c))$$

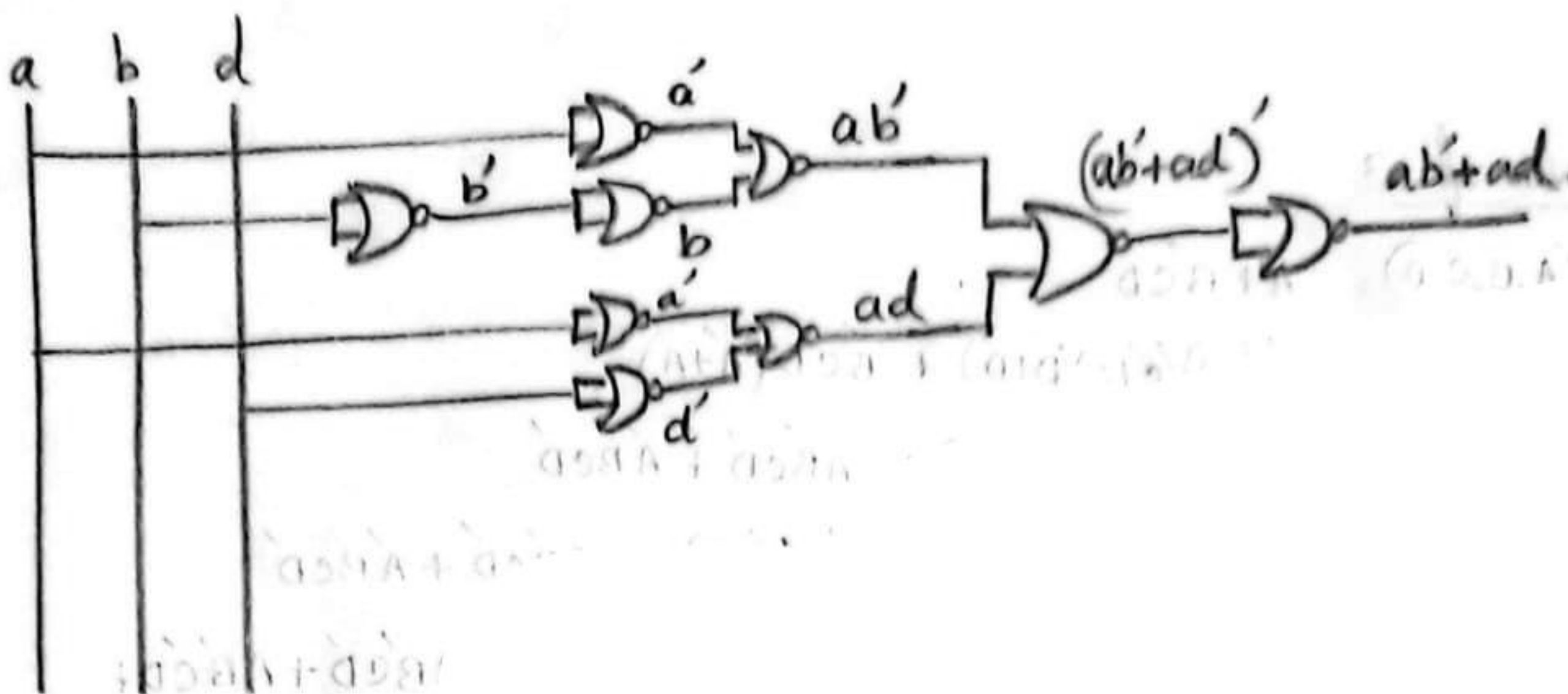
$$= ab'(c'+c) + abd$$

$$= ab' + abd$$

$$= a(b'+bd)$$

$$= a(b'+d) \quad [\because X + X'Y = X + Y]$$

$$= ab' + ad$$



Answer of 05

$$F(a, b, c, d) = \sum (5, 8, 9, 12, 15)$$

SOP Form, $(5, 8, 9, 12, 15)$

Binary

- 5 $\rightarrow$ 0101 $\rightarrow$ $a'b'cd$
- 8 $\rightarrow$ 1000 $\rightarrow$ $ab'c'd'$
- 9 $\rightarrow$ 1001 $\rightarrow$ $ab'c'd$
- 12 $\rightarrow$ 1100 $\rightarrow$ $abc'd'$
- 15 $\rightarrow$ 1111 $\rightarrow$ $abcd$

$$F = a'b'cd + ab'c'd' + ab'c'd + abc'd' + abcd$$

$$= a'b'cd + ab'c'(d' + d) + abc'd' + abcd$$

$$= a'b'cd + ab'c' + abc'd' + abcd$$

